

Problems 1, 2, 3 require you to have read the lecture notes or watched the lecture, but you should be able to attempt them on your own after that. They are important to attempt, in order to build a good understanding about the algorithms seen in class. (Problem 3 links this week's content to the topics covered in Chapter 4.)

Problem 4 asks you to use Matlab to compare algorithms, and gives you (some) practice using linear programming libraries.

Problem 5 is doable, with the exception of 5 c), which is more difficult.

Problem 6 is quite technical and on the difficult side, but is a great way to check you have understood the proofs from the lecture.

Problem 7 is interesting if you have time, and is very good (advanced) practice: it is not easy (and solutions are not provided)!

Warm-up

Problem 1. Check your understanding: recall their definitions, and summarise the key differences between an LP and an ILP.

Problem 2. Formulate MAX-CUT as an ILP.

- Give its LP relaxation, and suggest a randomised rounding strategy.
- Show that $y^* = (1, 1, \dots, 1)$ and $x^* = (1/2, 1/2, \dots, 1/2)$ is always an optimal solution to the LP relaxation.
- What does your rounding scheme become in this case?

Problem 3. Describe how to derandomise the 3/4-approximation algorithm for MAX-SAT given in class.

Problem solving

Problem 4. Consider the KNAPSACK problem, where the goal is to select a subset of n items that fit in the knapsack (which can only store total weight W) in order to maximise total value, where item i has value $v_i \geq 0$ and weight $w_i > 0$.

- Give the corresponding ILP.
- Provide the LP relaxation, which corresponds to the *Fractional Knapsack*.
- Solve the LP relaxation (using, e.g., Matlab with the function `linprog`) on the following set of 10 items, with weight limit $W = 20$: $(v_i, w_i) = (i^2, i)$, $1 \leq i \leq 10$. See how this changes as you vary W from 20 to 55.
- Compare to the solution obtained by the Greedy algorithm for Fractional Knapsack.

- e) Compare to the optimal solution of the ILP (for $W = 20$, then varying W as before), also obtained by solving the ILP (on Matlab, with the function `intlinprog`).

Problem 5. Suppose the instance of MAX-SAT has no negated “unit clause” (that is, either a clause has length at least 2, or it is a non-negated variable x_i). Instead of setting each variable to 1 independently with probability $1/2$ in the “obvious” randomised algorithm, do the analysis when this is done with some (fixed) probability $p > 1/2$.

- a) Show that this gives (in expectation) a $\min(p, 1 - p^2)$ -approximation.
- b) Optimise the choice of p to obtain the best approximation possible.
- c) (★) Show how to remove the “no negated unit clause” assumption: let $S \subseteq [n]$ be the set of variables such that both the unit clause $\neg x_i$ and the unit clause x_i exist in the instance ϕ , and $T \subseteq [n]$ be the set of variables for which only the unit clause $\neg x_i$ is in ϕ . Then consider the randomised rounding scheme with sets each variable i independently to 1 with probability p (as before) if $i \notin T$, and with probability $1 - p$ otherwise, where p is the value found in the previous subquestion. Show that $\text{opt}(\phi) \leq m - |S|$. Use this to conclude that $\mathbb{E}[\text{value}(\phi)] \geq p \cdot \text{opt}(\phi)$.
- d) Compare this with the $1 - 1/e$ approximation guarantee obtained by LP rounding in the lecture.

Problem 6. Show that one can also obtain (directly) an expected $\frac{3}{4}$ -approximation to MAX-SAT by using only randomised rounding: in Algorithm 20, instead of having $x_i \sim \text{Bern}(y_i^*)$ (independently), we will set them independently to 1 with probability

$$p_i := f(y_i^*),$$

where $f: [0, 1] \rightarrow [0, 1]$ is any function such that $1 - \frac{1}{4^x} \leq f(x) \leq \frac{1}{4^{1-x}}$.

- a) Draw the plot of both upper and lower bounds on f , to see what the conditions look like (and that such functions f do exist).
- b) In what follows, we fix any such function f . With the notation of Theorem 48, show that, for any $1 \leq j \leq m$,

$$\Pr[C_j \text{ not satisfied}] \leq \frac{1}{4^{z_j^*}}$$

- c) Deduce that, for any $1 \leq j \leq m$,

$$\Pr[C_j \text{ satisfied}] \geq \frac{3}{4} z_j^*$$

Hint: use concavity.

- d) Conclude.

Advanced

Problem 7. Show that one can also obtain (directly) an expected $\frac{3}{4}$ -approximation to MAX-SAT by using only randomised rounding with a *linear* function of y_i^* : in Algorithm 20, instead of having $x_i \sim \text{Bern}(y_i^*)$ (independently), set them independently to 1 with probability

$$p_i := \frac{y_i^*}{2} + \frac{1}{4}.$$